# COMP 018: Database Administration — Activity 6 Progress Report

*Database Backup Planning, Verification, Restoration, and Disaster Recovery Simulation*

## Project Identification & Metadata

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|---|---|
| **Course** | COMP 018 - Database Administration |
| **Evaluating Group** | Group 4: Olivarez, Abangan, Pocsidio, Infantado, and Bartolata |
| **Evaluating Group Repository** | https://github.com/Nycri/brewpos |
| **Target Project Monitored** | CampusSafe: Campus Hazard & Safety Reporting System |
| **Target Project Group** | Group 7 (Alba, Arreza, Orpesa, Robiso, Santos) |
| **Target Project Repository** | https://github.com/jeffersonorpesa2-ops/grp07-campussafe |
| **Primary Database** | grp07_campussafedb (PostgreSQL 18.6 on x86_64 Windows, compiled by MSVC) |
| **Restored Test Database** | grp07_campussafedb_restore_test |

## ACTIVITY 6: Database Backup, Restoration, and Recovery

### 3.1 Backup Planning & Architecture

The disaster recovery plan for the CampusSafe database targets all nine public relational tables, sequences, primary keys, foreign keys, unique constraints, and privilege definitions. This scope ensures data integrity across user profiles, location entries, hazard categories, incident reports, work dispatches, status logs, and file attachments.

The selected backup mechanism utilizes PostgreSQL custom compressed archive format via dedicated backup role credentials. Archive files are stored within a secure directory path separate from application web directories and version control repositories. Recovery parameters establish a Recovery Point Objective of 24 hours and a Recovery Time Objective of 2 hours, aligning with typical campus reporting activity. The retention strategy maintains seven daily backups alongside four weekly snapshots to support data retention policies.

Because database login accounts exist at the PostgreSQL cluster level rather than inside single-database dumps, disaster recovery procedures specify that access-control role initialization scripts must be executed on new server instances prior to restoring database contents.

## 3.2 Backup Creation, Checksum & Archive Inspection

Logical backup creation was executed using the designated backup account, producing a compressed custom format dump file sized at 47,711 bytes. Cryptographic verification computed a SHA-256 hash value of 73DC76542E218D4046B236C72A7FF58019862A38AC5DEF5FAE75D59D77FB99B2, establishing a baseline for integrity verification.

Inspection of the archive table of contents verified 118 catalog entries, confirming the inclusion of all nine database tables, sequence objects, primary key definitions, check constraints, indexes, and object privilege grants. To ensure full recoverability beyond archive listing checks, a complete restoration test was conducted in an isolated test database.

## 3.3 Restoration into Separate Test Database

An isolated test database named grp07_campussafedb_restore_test was created on the local server instance. The custom archive dump file was restored into this database using administrative utility tools. The restoration process completed successfully with zero error codes.

## 3.4 Post-Restore Verification Suite

Post-restoration verification confirmed that all nine relational tables were recreated with proper schema structure under administrative ownership. Record count validation across all tables demonstrated complete data parity with the source database, matching five role records, six user entries, six hazard categories, ten campus locations, seventeen hazard reports, twelve assignments, sixty status history logs, nine maintenance actions, and five attachment records.

Access control verification on the restored test database confirmed that privilege definitions transferred accurately. Queries executed under the runtime user account successfully returned hazard report records, while query attempts by the read-only auditor account on user credential tables were blocked by the database security layer.

## 3.5 Controlled Recovery Simulation

A controlled data recovery simulation was performed to test incident response procedures. A disposable hazard report record was inserted into the restored test database to establish a baseline state. A data loss event was then simulated by removing the record from the database, and queries confirmed its absence.

Targeted record restoration was executed to reinsert the missing data row without affecting existing database entries. Post-recovery verification confirmed that the record was fully restored with its original attributes, locations, and descriptions intact. Following verification, the test record was removed to clean up the test environment.

## 3.6 Limitations, Evaluation & Production Recommendations

Restoration procedures on fresh server installations require pre-executing role definition scripts to establish user accounts before restoring database ACLs. For production operations, daily

automated backup generation should be scheduled using system task schedulers to archive database dumps with date-stamped filenames and retain snapshots according to policy. For environments requiring near-zero data loss tolerances, enabling continuous write-ahead log archiving will support point-in-time recovery capabilities.

## CLASS GOOGLE SHEETS TRACKER ENTRY (Activity 6)

| Column Header | Value |
|---|---|
| **Project** | Group 7 - CampusSafe: Campus Hazard & Safety Reporting System |
| **Database and backup file** | grp07_campussafedb, grp07_campussafedb_activity6.dump, Size: 47,711 bytes, SHA256: 73DC76542E218D4046B236C72A7FF58019862A38AC5DEF5FAE75D59D77FB99B2 |
| **Backup method** | pg_dump custom format (-Fc), compressed binary dump (2026-11-05 02:13:17) |
| **Restore test result** | Success (Restored into grp07_campussafedb_restore_test; all 9 tables, constraints, and data verified) |
| **Access control after restore** | (Activity 4 roles and permissions validated; campussafe_report denied reading app_user) |
| **Recovery simulation result** | Recovered (Simulated deleted test hazard report successfully restored from baseline state) |
| **Proposed RPO and RTO** | RPO: 24 hours, RTO: 2 hours |
| **Evidence folder link** | https://github.com/Nycri/brewpos/tree/main/activity-6/Evidence-act6 |
| **Feedback sent to owners** | (05/11/2026) |
| **PDF progress document** | Activity6_Progress_Report_Group7.pdf |
| **Last updated** | 2026-11-05 |

# AI USAGE DECLARATION

Google Gemini 3.8 Flash was utilized during project evaluation to inspect database schema definitions, evaluate relational hierarchy structures, formulate role verification test scenarios, analyze database maintenance operations and query performance execution plans, verify backup creation and restoration procedures, and structure technical report documentation. All database statements, configuration checks, and verification procedures were directly executed and validated against PostgreSQL 18.6 within the local database environment.